Logic Block

U3E
TLV1824
Vsupply12
C5 10uf
C6 100nf
GND

5V supply
5kw level
R19 10k
R21 76.8k
R17 3296W-1-103LF
R25 84.5k
R18 10k
U3A TLV1824
R20 1M
5kw level demand reached
R30

R31 309k
R22 3296W-1-103LF
R41 20k
R24 1M
R23 10k
U3B TLV1824
signal from 5kw level being recieved at all?

R3 4.99k
R27 3296W-1-103LF
R47 110k
R29 10k
R28 10k
1M
U3C TLV1824
Short some where in the 5kw level signal
U3D TLV1824
U4E TLV1824
C7 10uf
C8 100nf
Brake Sensor
R34 10k

R48 place holder
R26 3296W-1-103LF
R49 place holder
R37 10k
U4A TLV1824
R42 1M
Brake Sensor level being reached

R50 309k
R33 3296W-1-103LF
R51 20k
R38 10k
U4B TLV1824
R40 1M
signal from Brake Sensor being recieved at all?

R52 4.99k
R32 3296W-1-103LF
R53 110k
R36 10k
R35 10k
U4C TLV1824
R39 1M
Short some where in the Brake Sensor signal
U4D TLV1824

data sheet: https://www.ti.com/lit/ds/symlink/tlv1824-q1.pdf?ts=1791174079110&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FTLV1824-Q1

IC Configure tool:https://beta-tools.analog.com/timerblox/LTC6994
delay amount .470s

5V supply
R1 523k
C11 100nf
Brake_and_5KW
R2 1000k
R3 715K
U1 LTC6994xS6-2
DIV IN SET GND V+ OUT
TP3 TestPoint
Brake_and_5KW_After_Delay

Delay block

U2
TPS7A2501DRVR
D1 SS14
Vsupply12
C10 10uf
C12 100nf
IN EN OUT PG NC GND Thermal_Pad
C13 10uf
5V supply
R5 309K
R4 102k

12v to 5v LDO linear regulator

data sheet : https://www.ti.com/lit/ds/symlink/tps7a25.pdf?HQS=dis-dk-null-digikeymode-dsf-pf-null-wwe&ts=1791170323719

R44 10k
5kw level demand reached
Brake Sensor level being reached
TP2 TestPoint
Brake_and_5KW
Low When conditions arent met
High when conditions are met->car turns off

R45 10k
signal from 5kw level being recieved at all?
Short some where in the 5kw level signal
signal from Brake Sensor being recieved at all?
Short some where in the Brake Sensor signal
TP1 TestPoint
No fault in sensor wiring
Reads High when everything is reading good
Reads Low when/ if faults in sensors

R46 10k
No fault in sensor wiring
C9 1uf
Latch Input Sensor after 10ms

Q2_GATE
U6
MMSD4148T1G
ANODE CATHODE
RESET_N
Vsupply12
C14 2.2uf
R9 22k
Q3 MMBT3904
D3 1N4148W
R10 100k

Brake_and_5KW_After_Delay
R8 1.5k
Latch Input Sensor after 10ms
Latching Circuit
5V supply
C15 10uf
C16 100nf
U5
SN74HC132PWR
1A 1B 1Y 2A 2B 2Y GND VCC 4B 4A 4Y 3B 3A 3Y
Fault_Latch
Fault_Latch
Set_Net
Tripped
OK
OK
Tripped
RESET_N
Set_Net
R14 10k
C17 100nf
RESET

OK
R12 10k
R13 100k
Q2_GATE
Q2 2N7002K
D4 SS14
K1 G5Q-1A-EU DC12
Vsupply12
SDC_IN
SDC_OUT

Connectors
Current Sensor -
Current Sensor +
GND
5V supply
SDC_IN
SDC_OUT
J2 436500215
Brake Sensor
Vsupply12
5kw level
J3 430450612
TP4 TestPoint

H3 MountingHole
HMountingHole_2.2mm_M2 MountingHole
H2 MountingHole
H1 MountingHole

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